

Solution of assignment#2

Heating Ammonia at Constant Pressure

A vertical piston–cylinder assembly containing 0.1 lb of ammonia, initially a saturated vapor, is placed on a hot plate. Due to the weight of the piston and the surrounding atmospheric pressure, the pressure of the ammonia is 20 lbf/in.² Heating occurs slowly, and the ammonia expands at constant pressure until the final temperature is 77°F. Show the initial and final states on T – v and p – v diagrams, and determine

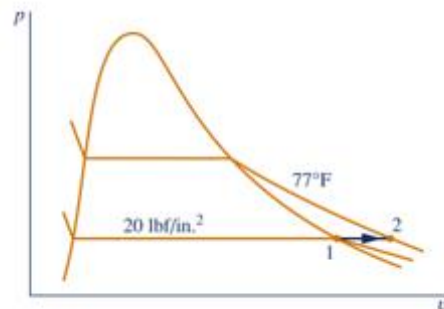
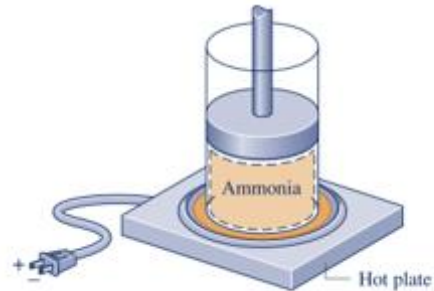
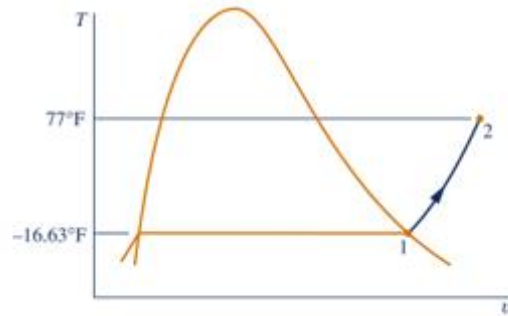
- (a) the volume occupied by the ammonia at each end state, in ft³.
- (b) the work for the process, in Btu.

SOLUTION

Known: Ammonia is heated at constant pressure in a vertical piston–cylinder assembly from the saturated vapor state to a known final temperature.

Find: Show the initial and final states on T – v and p – v diagrams, and determine the volume at each end state and the work for the process.

Schematic and Given Data:



Engineering Model:

1. The ammonia is a closed system.
2. States 1 and 2 are equilibrium states.
3. The process occurs at constant pressure.
4. The piston is the only work mode.

Analysis: The initial state is a saturated vapor condition at 20 lbf/in.² Since the process occurs at constant pressure, the final state is in the superheated vapor region and is fixed by $p_2 = 20$ lbf/in.² and $T_2 = 77^\circ\text{F}$. The initial and final states are shown on the T - v and p - v diagrams above.

(a) The volumes occupied by the ammonia at states 1 and 2 are obtained using the given mass and the respective specific volumes. From Table A-15E at $p_1 = 20$ lbf/in.², and corresponding to *Sat.* in the temperature column, we get $v_1 = v_g = 13.497$ ft³/lb. Thus,

$$\begin{aligned} V_1 &= mv_1 = (0.1 \text{ lb})(13.497 \text{ ft}^3/\text{lb}) \\ &= 1.35 \text{ ft}^3 \end{aligned}$$

Interpolating in Table A-15E at $p_2 = 20$ lbf/in.² and $T_2 = 77^\circ\text{F}$, we get $v_2 = 16.7$ ft³/lb. Thus,

$$V_2 = mv_2 = (0.1 \text{ lb})(16.7 \text{ ft}^3/\text{lb}) = 1.67 \text{ ft}^3$$

(b) In this case, the work can be evaluated using Eq. 2.17. Since the pressure is constant

$$W = \int_{V_1}^{V_2} p \, dV = p(V_2 - V_1)$$

Inserting values

$$\begin{aligned} W &= (20 \text{ lbf/in.}^2)(1.67 - 1.35) \text{ ft}^3 \left| \frac{144 \text{ in.}^2}{1 \text{ ft}^2} \right| \left| \frac{1 \text{ Btu}}{778 \text{ ft} \cdot \text{lbf}} \right| \\ &= 1.18 \text{ Btu} \end{aligned}$$